

Skills Summary

Groups, Arrays, and Products

Use this reference while completing your worksheet or when studying.

1. Adding the same number again and again

When the *same* number is added over and over, count how many times it appears — that is the number of groups. **groups** $\times$ **size of a group** = **total**.

$4 + 4 + 4$ is three groups of four, so it is 3×4 .

Example: Write $4 + 4 + 4$ as a multiplication.

Step 1. Every group is the same size, so counting the groups is quicker than counting everything one by one.

Step 2. There are 3 fours, so this is 3×4 : count 4, 8, 12.

$$3 \times 4 = 12$$

Try it: Write $5 + 5 + 5 + 5$ as a multiplication and find the total.

check: $4 \times 5 = 20$

2. Reading an array: rows and row size

An array has rows that all hold the same number of things. Read it as **rows** $\times$ **how many in each row**.

Turn the array sideways and you read the same picture the other way, so 3×5 and 5×3 give the same total.

Example: An array has 3 rows with 5 dots in each row.

Step 1. Counting all the dots works but is slow — the rows are equal, so count rows and row size instead.

Step 2. 3 rows of 5 is 3×5 : count by fives, 5, 10, 15.

$$3 \times 5 = 15$$

Try it: An array has 4 rows with 6 tiles in each row. How many tiles?

check: $4 \times 6 = 24$

3. Turning a product back into an addition

5×2 is read as *5 groups of 2*. The **second** number is what repeats, and the **first** number says how many times.

So $5 \times 2 = 2 + 2 + 2 + 2 + 2$.

Example: Write 5×2 as a repeated addition.

Step 1. Say the multiplication out loud as “five groups of two” — that sentence tells you which number to write down over and over.

Step 2. Write 2 five times: $2 + 2 + 2 + 2 + 2$, and count by twos: 2, 4, 6, 8, 10.

$$5 \times 2 = 10$$

Try it: Write 3×6 as a repeated addition and find the total.

check: $9 + 9 = 18$

4. Finding the missing factor

$3 \times \underline{\quad} = 18$ asks: three of *what* makes 18? Skip-count by the number you know and stop when you reach the total — the number of counts you made is the missing factor.

Example: Fill in $3 \times \underline{\quad} = 18$.

Step 1. One factor and the total are known, so skip-counting by the known factor will run straight into the answer.

Step 2. Count by threes: 3, 6, 9, 12, 15, 18 — that is six threes.

The missing factor is **6**.

Try it: Fill in $\underline{\quad} \times 4 = 28$.

check: 7

(!) Watch out: 4×6 is not $4 + 6$. The two numbers have different jobs — one counts the groups, the other says how big each group is.